```
C:\Users\user\AppData\Local\Programs\Python\Python313\python.exe C:\Users\user\tsc\core\geo_cluster.py
C:\Users\user\tsc\core\geo_cluster.py:19: DeprecationWarning: `scipy.special.sph_harm` is deprecated as of SciPy 1.15.0 and will be removed in SciPy 1.17.0. Please use `scipy.special.sph_harm_y` instead.
  tm = scipy.special.sph_harm(m, l, ph, th).real
C:\Users\user\tsc\core\geo_cluster.py:17: DeprecationWarning: `scipy.special.sph_harm` is deprecated as of SciPy 1.15.0 and will be removed in SciPy 1.17.0. Please use `scipy.special.sph_harm_y` instead.
  tm = scipy.special.sph_harm(-m, l, ph, th).imag
[17:49:03] UFFTYPER: Unrecognized charge state for atom: 8
[17:49:28] UFFTYPER: Unrecognized atom type: Ca+2 (0)
[17:59:12] UFFTYPER: Unrecognized charge state for atom: 1
[17:59:12] UFFTYPER: Unrecognized charge state for atom: 2
[17:59:13] UFFTYPER: Unrecognized charge state for atom: 6
[17:59:13] UFFTYPER: Unrecognized charge state for atom: 6
[18:01:40] UFFTYPER: Unrecognized charge state for atom: 0
[18:01:40] UFFTYPER: Unrecognized atom type: Zn+2 (0)
C:\Users\user\tsc\core\geo_cluster.py:44: RuntimeWarning: invalid value encountered in scalar divide
  th[i] = math.acos(coords[i, 2] / r[i])
[18:02:20] UFFTYPER: Unrecognized atom type: As2+3 (1)
[18:02:20] UFFTYPER: Unrecognized atom type: As2+3 (1)
[18:02:21] UFFTYPER: Unrecognized atom type: S_6+6 (32)
[18:02:21] UFFTYPER: Unrecognized atom type: S_6+6 (32)
[18:03:17] UFFTYPER: Unrecognized charge state for atom: 17
[18:03:22] UFFTYPER: Unrecognized charge state for atom: 1
[18:04:07] UFFTYPER: Unrecognized charge state for atom: 1
[18:06:30] UFFTYPER: Unrecognized charge state for atom: 1
[18:11:07] UFFTYPER: Unrecognized charge state for atom: 1
[18:16:11] UFFTYPER: Unrecognized charge state for atom: 54
[18:16:40] UFFTYPER: Unrecognized charge state for atom: 1
[18:16:53] UFFTYPER: Unrecognized charge state for atom: 1
[18:17:45] UFFTYPER: Unrecognized charge state for atom: 3
[18:17:59] UFFTYPER: Unrecognized charge state for atom: 4
[18:19:24] UFFTYPER: Unrecognized atom type: S_6+6 (12)
[18:19:24] UFFTYPER: Unrecognized atom type: S_6+6 (12)
[18:20:39] UFFTYPER: Unrecognized charge state for atom: 1
[18:26:11] UFFTYPER: Unrecognized atom type: Sr (0)
[18:26:11] UFFTYPER: Unrecognized atom type: Sr (8)
[18:27:06] UFFTYPER: Unrecognized charge state for atom: 23
[18:27:13] UFFTYPER: Unrecognized charge state for atom: 20
[18:29:04] UFFTYPER: Unrecognized charge state for atom: 5
[18:30:44] UFFTYPER: Unrecognized charge state for atom: 12
[18:30:45] UFFTYPER: Unrecognized charge state for atom: 3
[18:30:49] UFFTYPER: Unrecognized charge state for atom: 14
```

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41 [18:30:55] UFFTYPER: Unrecognized charge state for atom: 10
42 [WARN] Failed SH for mol_8891: math domain error
43 [18:30:56] UFFTYPER: Unrecognized charge state for atom: 1
44 [18:32:10] UFFTYPER: Unrecognized charge state for atom: 8
45 [18:32:53] UFFTYPER: Unrecognized charge state for atom: 8
46 [18:33:08] UFFTYPER: Unrecognized atom type: Ca+2 (0)
47 [18:33:34] UFFTYPER: Unrecognized charge state for atom: 1
48 [18:34:54] UFFTYPER: Unrecognized atom type: S_6+6 (9)
49 [18:34:54] UFFTYPER: Unrecognized atom type: S_6+6 (9)
50 [18:41:52] UFFTYPER: Unrecognized charge state for atom: 1
51 [18:41:52] UFFTYPER: Unrecognized charge state for atom: 2
52 [18:41:57] UFFTYPER: Unrecognized charge state for atom: 21
53 [18:44:07] UFFTYPER: Unrecognized atom type: S_6+6 (10)
54 [18:44:07] UFFTYPER: Unrecognized atom type: S_6+6 (10)
55 [18:44:38] UFFTYPER: Warning: hybridization set to SP3 for atom
    29
56 [18:44:38] UFFTYPER: Warning: hybridization set to SP3 for atom
    29
57 [18:44:38] UFFTYPER: Warning: hybridization set to SP3 for atom
    29
58 [18:45:52] UFFTYPER: Unrecognized atom type: S_6+6 (32)
59 [18:45:52] UFFTYPER: Unrecognized atom type: S_6+6 (32)
60 [18:47:00] UFFTYPER: Unrecognized charge state for atom: 17
61 [19:30:22] UFFTYPER: Unrecognized charge state for atom: 20
62 [19:31:10] UFFTYPER: Unrecognized charge state for atom: 21
63 [19:31:33] UFFTYPER: Warning: hybridization set to SP3 for atom
    23
64 [19:31:33] UFFTYPER: Warning: hybridization set to SP3 for atom
    23
65 [19:31:33] UFFTYPER: Warning: hybridization set to SP3 for atom
    23
66 [19:33:40] UFFTYPER: Unrecognized atom type: S_6+6 (11)
67 [19:33:40] UFFTYPER: Unrecognized atom type: S_6+6 (11)
68 [19:35:30] UFFTYPER: Warning: hybridization set to SP3 for atom
    27
69 [19:35:30] UFFTYPER: Warning: hybridization set to SP3 for atom
    27
70 [19:35:30] UFFTYPER: Warning: hybridization set to SP3 for atom
    27
71 [19:40:07] UFFTYPER: Unrecognized charge state for atom: 10
72 [19:44:03] UFFTYPER: Unrecognized charge state for atom: 1
73 [19:55:01] UFFTYPER: Unrecognized charge state for atom: 19
74 [19:55:18] UFFTYPER: Unrecognized charge state for atom: 3
75 [19:56:08] UFFTYPER: Unrecognized charge state for atom: 4
76 [19:59:11] UFFTYPER: Unrecognized atom type: S_6+6 (12)
77 [19:59:11] UFFTYPER: Unrecognized atom type: S_6+6 (12)
78 [20:12:32] UFFTYPER: Unrecognized charge state for atom: 23
79 [20:12:49] UFFTYPER: Unrecognized charge state for atom: 20
```

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 80 [20:13:20] UFFTYPER: Warning: hybridization set to SP3 for
    atom 0
 81 [20:13:20] UFFTYPER: Unrecognized charge state for atom: 0
 82 [20:39:15] UFFTYPER: Unrecognized charge state for atom: 5
 83 [20:42:51] UFFTYPER: Unrecognized atom type: Ca+2 (0)
 84 [20:42:52] UFFTYPER: Unrecognized charge state for atom: 12
 85 [20:42:53] UFFTYPER: Unrecognized charge state for atom: 3
 86 [20:43:02] UFFTYPER: Unrecognized charge state for atom: 14
 87 [WARN] Failed SH for mol_18086: math domain error
 88 [20:43:34] UFFTYPER: Unrecognized charge state for atom: 21
 89 [20:44:28] UFFTYPER: Unrecognized charge state for atom: 8
 90 [20:46:06] UFFTYPER: Unrecognized atom type: S_6+6 (9)
 91 [20:46:06] UFFTYPER: Unrecognized atom type: S_6+6 (9)
 92 [20:50:36] UFFTYPER: Unrecognized atom type: S_6+6 (10)
 93 [20:50:36] UFFTYPER: Unrecognized atom type: S_6+6 (10)
 94 [20:52:04] UFFTYPER: Unrecognized atom type: S_6+6 (32)
 95 [20:52:05] UFFTYPER: Unrecognized atom type: S_6+6 (32)
 96 [20:55:03] UFFTYPER: Unrecognized atom type: S_6+6 (11)
 97 [20:55:03] UFFTYPER: Unrecognized atom type: S_6+6 (11)
 98 [21:02:54] UFFTYPER: Unrecognized atom type: S_6+6 (12)
 99 [21:02:54] UFFTYPER: Unrecognized atom type: S_6+6 (12)
100 [21:06:49] UFFTYPER: Unrecognized atom type: S_6+6 (9)
101 [21:06:49] UFFTYPER: Unrecognized atom type: S_6+6 (9)
102 [21:11:05] UFFTYPER: Unrecognized atom type: S_6+6 (10)
103 [21:11:05] UFFTYPER: Unrecognized atom type: S_6+6 (10)
104 [21:12:31] UFFTYPER: Unrecognized atom type: S_6+6 (32)
105 [21:12:31] UFFTYPER: Unrecognized atom type: S_6+6 (32)
106 [21:15:32] UFFTYPER: Unrecognized atom type: S_6+6 (11)
107 [21:15:32] UFFTYPER: Unrecognized atom type: S_6+6 (11)
108 [21:24:38] UFFTYPER: Unrecognized atom type: S_6+6 (12)
109 [21:24:39] UFFTYPER: Unrecognized atom type: S_6+6 (12)
110 [21:29:11] UFFTYPER: Unrecognized atom type: S_6+6 (9)
111 [21:29:11] UFFTYPER: Unrecognized atom type: S_6+6 (9)
112 [21:35:07] UFFTYPER: Unrecognized charge state for atom: 1
113 [21:35:07] UFFTYPER: Unrecognized charge state for atom: 2
114 [21:37:19] UFFTYPER: Unrecognized atom type: S_6+6 (10)
115 [21:37:19] UFFTYPER: Unrecognized atom type: S_6+6 (10)
116 [21:39:12] UFFTYPER: Unrecognized atom type: S_6+6 (32)
117 [21:39:12] UFFTYPER: Unrecognized atom type: S_6+6 (32)
118 [21:45:01] UFFTYPER: Unrecognized atom type: S_6+6 (11)
119 [21:45:01] UFFTYPER: Unrecognized atom type: S_6+6 (11)
120 [21:56:08] UFFTYPER: Unrecognized charge state for atom: 4
121 [21:56:44] UFFTYPER: Unrecognized atom type: S_6+6 (12)
122 [21:56:44] UFFTYPER: Unrecognized atom type: S_6+6 (12)
123 [21:59:55] UFFTYPER: Unrecognized charge state for atom: 14
124 [22:00:06] UFFTYPER: Unrecognized atom type: S_6+6 (9)
125 [22:00:06] UFFTYPER: Unrecognized atom type: S_6+6 (9)
126 [22:00:38] UFFTYPER: Unrecognized atom type: S_6+6 (10)
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127 [22:00:38] UFFTYPER: Unrecognized atom type: S_6+6 (10)
128 [22:00:46] UFFTYPER: Unrecognized atom type: S_6+6 (32)
129 [22:00:47] UFFTYPER: Unrecognized atom type: S_6+6 (32)
130 [22:01:14] UFFTYPER: Unrecognized atom type: S_6+6 (11)
131 [22:01:14] UFFTYPER: Unrecognized atom type: S_6+6 (11)
132 [22:02:37] UFFTYPER: Unrecognized atom type: S_6+6 (12)
133 [22:02:37] UFFTYPER: Unrecognized atom type: S_6+6 (12)
134 [22:04:05] UFFTYPER: Unrecognized charge state for atom: 8
135 [22:04:18] UFFTYPER: Unrecognized atom type: Ca+2 (0)
136 [22:04:54] UFFTYPER: Unrecognized charge state for atom: 1
137 [22:05:33] UFFTYPER: Unrecognized atom type: S_6+6 (9)
138 [22:05:33] UFFTYPER: Unrecognized atom type: S_6+6 (9)
139 [22:07:03] UFFTYPER: Unrecognized charge state for atom: 1
140 [22:07:03] UFFTYPER: Unrecognized charge state for atom: 2
141 [22:07:03] UFFTYPER: Unrecognized charge state for atom: 6
142 [22:07:03] UFFTYPER: Unrecognized charge state for atom: 6
143 [22:07:51] UFFTYPER: Unrecognized atom type: S_6+6 (10)
144 [22:07:51] UFFTYPER: Unrecognized atom type: S_6+6 (10)
145 [22:08:11] UFFTYPER: Unrecognized charge state for atom: 0
146 [22:08:11] UFFTYPER: Unrecognized atom type: Zn+2 (0)
147 [22:08:57] UFFTYPER: Unrecognized atom type: As2+3 (1)
148 [22:08:57] UFFTYPER: Unrecognized atom type: As2+3 (1)
149 [22:08:57] UFFTYPER: Unrecognized atom type: S_6+6 (32)
150 [22:08:57] UFFTYPER: Unrecognized atom type: S_6+6 (32)
151 [22:09:20] UFFTYPER: Unrecognized charge state for atom: 17
152 [22:09:52] UFFTYPER: Unrecognized charge state for atom: 20
153 [22:10:28] UFFTYPER: Unrecognized atom type: S_6+6 (11)
154 [22:10:28] UFFTYPER: Unrecognized atom type: S_6+6 (11)
155 [22:14:21] UFFTYPER: Unrecognized charge state for atom: 54
156 [22:14:33] UFFTYPER: Unrecognized charge state for atom: 1
157 [22:15:02] UFFTYPER: Unrecognized charge state for atom: 3
158 [22:15:18] UFFTYPER: Unrecognized charge state for atom: 4
159 [22:15:54] UFFTYPER: Unrecognized atom type: S_6+6 (12)
160 [22:15:54] UFFTYPER: Unrecognized atom type: S_6+6 (12)
161 [22:16:28] UFFTYPER: Unrecognized charge state for atom: 1
162 [22:19:30] UFFTYPER: Unrecognized atom type: Sr (0)
163 [22:19:30] UFFTYPER: Unrecognized atom type: Sr (8)
164 [22:20:18] UFFTYPER: Unrecognized charge state for atom: 23
165 [22:20:26] UFFTYPER: Unrecognized charge state for atom: 20
166 [22:20:32] UFFTYPER: Warning: hybridization set to SP3 for
    atom 0
167 [22:20:32] UFFTYPER: Unrecognized charge state for atom: 0
168 [22:22:59] UFFTYPER: Unrecognized charge state for atom: 5
169 [22:24:57] UFFTYPER: Unrecognized atom type: Ca+2 (0)
170 [22:24:59] UFFTYPER: Unrecognized charge state for atom: 12
171 [22:25:00] UFFTYPER: Unrecognized charge state for atom: 3
172 [22:25:03] UFFTYPER: Unrecognized charge state for atom: 14
173 [22:25:10] UFFTYPER: Unrecognized charge state for atom: 10
```

```
[WARN] Failed SH for mol_32870: math domain error
[22:25:11] UFFTYPER: Unrecognized charge state for atom: 1
[22:25:17] UFFTYPER: Unrecognized charge state for atom: 21
[22:26:27] UFFTYPER: Unrecognized charge state for atom: 8
[22:27:18] UFFTYPER: Unrecognized charge state for atom: 8
[22:27:30] UFFTYPER: Unrecognized atom type: Ca+2 (0)
[22:31:15] UFFTYPER: Unrecognized charge state for atom: 6
[22:31:15] UFFTYPER: Unrecognized charge state for atom: 6
[22:33:08] UFFTYPER: Unrecognized charge state for atom: 0
[22:33:08] UFFTYPER: Unrecognized atom type: Zn+2 (0)
[22:33:36] UFFTYPER: Unrecognized atom type: As2+3 (1)
[22:33:36] UFFTYPER: Unrecognized atom type: As2+3 (1)
[22:33:36] UFFTYPER: Unrecognized atom type: S_6+6 (32)
[22:33:36] UFFTYPER: Unrecognized atom type: S_6+6 (32)
[22:34:18] UFFTYPER: Unrecognized charge state for atom: 17
[22:34:22] UFFTYPER: Unrecognized charge state for atom: 1
[22:34:48] UFFTYPER: Unrecognized charge state for atom: 1
[22:36:12] UFFTYPER: Unrecognized charge state for atom: 1
[22:42:05] UFFTYPER: Unrecognized charge state for atom: 54
[22:42:27] UFFTYPER: Unrecognized charge state for atom: 1
[22:42:36] UFFTYPER: Unrecognized charge state for atom: 1
[22:43:13] UFFTYPER: Unrecognized charge state for atom: 3
[22:43:26] UFFTYPER: Unrecognized charge state for atom: 4
[22:44:16] UFFTYPER: Unrecognized atom type: S_6+6 (12)
[22:44:16] UFFTYPER: Unrecognized atom type: S_6+6 (12)
[22:45:02] UFFTYPER: Unrecognized charge state for atom: 1
[22:48:23] UFFTYPER: Unrecognized atom type: Sr (0)
[22:48:23] UFFTYPER: Unrecognized atom type: Sr (8)
[22:49:01] UFFTYPER: Unrecognized charge state for atom: 23
[22:49:06] UFFTYPER: Unrecognized charge state for atom: 20
[22:50:42] UFFTYPER: Unrecognized charge state for atom: 5
[22:52:07] UFFTYPER: Unrecognized charge state for atom: 12
[22:52:08] UFFTYPER: Unrecognized charge state for atom: 3
[22:52:11] UFFTYPER: Unrecognized charge state for atom: 14
[22:52:17] UFFTYPER: Unrecognized charge state for atom: 10
[WARN] Failed SH for mol_40040: math domain error
[22:52:18] UFFTYPER: Unrecognized charge state for atom: 1
[22:53:27] UFFTYPER: Unrecognized charge state for atom: 8
[22:59:54] UFFTYPER: Unrecognized atom type: S_6+6 (9)
[22:59:54] UFFTYPER: Unrecognized atom type: S_6+6 (9)
[23:00:51] UFFTYPER: Unrecognized atom type: S_6+6 (10)
[23:00:51] UFFTYPER: Unrecognized atom type: S_6+6 (10)
[23:01:06] UFFTYPER: Unrecognized atom type: S_6+6 (32)
[23:01:06] UFFTYPER: Unrecognized atom type: S_6+6 (32)
[23:01:50] UFFTYPER: Unrecognized atom type: S_6+6 (11)
[23:01:50] UFFTYPER: Unrecognized atom type: S_6+6 (11)
[23:03:34] UFFTYPER: Unrecognized charge state for atom: 4
[23:03:49] UFFTYPER: Unrecognized atom type: S_6+6 (12)
```

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[23:03:49] UFFTYPER: Unrecognized atom type: S_6+6 (12)
[23:04:43] UFFTYPER: Unrecognized charge state for atom: 8
[23:06:11] UFFTYPER: Unrecognized atom type: S_6+6 (9)
[23:06:11] UFFTYPER: Unrecognized atom type: S_6+6 (9)
[23:07:52] UFFTYPER: Unrecognized charge state for atom: 1
[23:07:55] UFFTYPER: Unrecognized charge state for atom: 5
[23:09:23] UFFTYPER: Unrecognized charge state for atom: 6
[23:09:23] UFFTYPER: Unrecognized charge state for atom: 6
[23:10:22] UFFTYPER: Unrecognized charge state for atom: 8
[23:10:27] UFFTYPER: Unrecognized atom type: S_6+6 (10)
[23:10:27] UFFTYPER: Unrecognized atom type: S_6+6 (10)
[23:10:54] UFFTYPER: Unrecognized charge state for atom: 0
[23:10:54] UFFTYPER: Unrecognized atom type: Zn+2 (0)
[23:11:18] UFFTYPER: Unrecognized atom type: As2+3 (1)
[23:11:18] UFFTYPER: Unrecognized atom type: As2+3 (1)
[23:11:19] UFFTYPER: Unrecognized atom type: S_6+6 (32)
[23:11:19] UFFTYPER: Unrecognized atom type: S_6+6 (32)
[23:13:49] UFFTYPER: Unrecognized atom type: S_6+6 (11)
[23:13:49] UFFTYPER: Unrecognized atom type: S_6+6 (11)
[23:15:36] UFFTYPER: Unrecognized charge state for atom: 1
[23:19:15] UFFTYPER: Unrecognized charge state for atom: 54
[23:19:29] UFFTYPER: Unrecognized charge state for atom: 1
[23:20:11] UFFTYPER: Unrecognized charge state for atom: 4
[23:20:41] UFFTYPER: Unrecognized charge state for atom: 5
[23:21:06] UFFTYPER: Unrecognized atom type: S_6+6 (12)
[23:21:06] UFFTYPER: Unrecognized atom type: S_6+6 (12)
[23:24:23] UFFTYPER: Unrecognized atom type: Sr (0)
[23:24:23] UFFTYPER: Unrecognized atom type: Sr (8)
[23:25:02] UFFTYPER: Unrecognized charge state for atom: 20
[23:26:17] UFFTYPER: Unrecognized charge state for atom: 5
[23:27:04] UFFTYPER: Unrecognized charge state for atom: 12
[23:27:06] UFFTYPER: Unrecognized charge state for atom: 14
[23:27:09] UFFTYPER: Unrecognized charge state for atom: 10
[23:27:10] UFFTYPER: Unrecognized charge state for atom: 1
[23:28:04] UFFTYPER: Unrecognized charge state for atom: 8
Traceback (most recent call last):
  File "C:\Users\user\tsc\core\geo_cluster.py", line 124, in <
module>
    result_df = cluster_smiles_csv("data/full_sdfs.csv",
cluster_count=5, max_mols=100)
  File "C:\Users\user\tsc\core\geo_cluster.py", line 106, in
cluster_smiles_csv
    labels = kmeans.fit_predict(features_scaled)
  File "C:\Users\user\AppData\Local\Programs\Python\Python313\
Lib\site-packages\sklearn\cluster\_kmeans.py", line 1064, in
fit_predict
    return self.fit(X, sample_weight=sample_weight).labels_
           ~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
  File "C:\Users\user\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\base.py", line 1389, in wrapper
    return fit_method(estimator, *args, **kwargs)
  File "C:\Users\user\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\cluster\_kmeans.py", line 1454, in fit
    X = validate_data(
        self,
    ...<5 lines>...
        accept_large_sparse=False,
    )
  File "C:\Users\user\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\utils\validation.py", line 2944, in validate_data
    out = check_array(X, input_name="X", **check_params)
  File "C:\Users\user\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\utils\validation.py", line 1107, in check_array
    _assert_all_finite(
    ~~~~~~~~~~~~~~~~~~^
        array,
        ^^^^^^
    ...<2 lines>...
        allow_nan=ensure_all_finite == "allow-nan",
        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "C:\Users\user\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\utils\validation.py", line 120, in _assert_all_finite
    _assert_all_finite_element_wise(
    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^
        X,
        ^^
    ...<4 lines>...
        input_name=input_name,
        ^^^^^^^^^^^^^^^^^^^^^^
    )
    ^
  File "C:\Users\user\AppData\Local\Programs\Python\Python313\Lib\site-packages\sklearn\utils\validation.py", line 169, in _assert_all_finite_element_wise
    raise ValueError(msg_err)
ValueError: Input X contains NaN.
KMeans does not accept missing values encoded as NaN natively. For supervised learning, you might want to consider sklearn.ensemble.HistGradientBoostingClassifier and Regressor which accept missing values encoded as NaNs natively. Alternatively
```

```
298 , it is possible to preprocess the data, for instance by using
     an imputer transformer in a pipeline or drop samples with
    missing values. See https://scikit-learn.org/stable/modules/
    impute.html You can find a list of all estimators that handle
    NaN values at the following page: https://scikit-learn.org/
    stable/modules/impute.html#estimators-that-handle-nan-values
299
300 Process finished with exit code 1
301
```